

Tuong Luu

(226) 791-2248 | tm2luu@uwaterloo.ca | [linkedin.com/in/luut189](https://www.linkedin.com/in/luut189) | github.com/luut189 | kyzel.dev/

EDUCATION

University of Waterloo

Candidate for Bachelor of Honours Computational Mathematics

Waterloo, Ontario

Sep 2023 - May 2028

TECHNICAL SKILLS

Languages: Java, Python, FastAPI, C/C++, JavaScript/TypeScript, Node.js, Express, React, Next.js

Databases: PostgreSQL, MongoDB, GCP (GKE, Secret Manager)

Tools: Git, Docker, Kubernetes (K8s), Linux, Bash, VS Code, Postman

Certifications: Azure Fundamentals, Azure AI Fundamentals

EXPERIENCE

Software Engineer Intern

Carta

Kitchener, Ontario

May. 2026 – Present

- Authored **520+** dbt tests across **100+** Snowflake models in the customer-facing fund administration data layer, surfacing **890K+** rows of latent data-quality defects and closing a coverage gap that had let a silent production outage pass undetected
- Shipped self-service Snowflake token authentication end-to-end across Terraform, Django, and React, replacing a manual per-customer provisioning process with a feature-flagged rollout covering **79** files and **150+** tests
- Designed and delivered the migration of Carta's public API documentation to a code-first **DocuSaurus** site, building a manifest-driven content pipeline and Kong/Helm routing across environments to replace a UI-authored third-party platform
- Built regression test suites for AI agent skills using the internal LLM evaluation framework, adding query-shape and routing assertions and patching a framework-level matcher bug that caused intermittent false failures
- Eliminated recurring on-call pages by root-causing production test failures to upstream data gaps and API gateway path misconfiguration, cutting alert noise from **255K+** to **2.5K** flagged rows

Software Developer Intern

Lynkr Inc

Toronto, Ontario

Sep. 2025 – Dec. 2025

- Collaborated with engineers to develop a FastAPI and Next.js application with AI agent orchestration, generating **\$100K+ CAD** in revenue and supporting **500+ beta users**
- Migrated the database seeder to a Kubernetes job, reducing deployment time on GKE by **40%**
- Architected multi-organization support with tenant isolation and RBAC, enabling secure segregation for and independent configuration per tenant
- Built account security features with MFA using TOTP and Twilio, along with JWT session controls, cutting unauthorized access by **70%** and protecting **500+** active sessions
- Developed an audit logging system tracking all user actions, admin changes, and security events, supporting compliance for **100% of critical operations**

WE Accelerate Program - Azure and Artificial Intelligence

University of Waterloo

Waterloo, Ontario

Jan. 2025 – Apr. 2025

- Led a team of 5 to design an AI-based solution preventing financial fraud against elders.
- Developed a comprehensive project planning tool and pipeline sequence wireframe, visualizing each step and aligning deliverables with industry best practices.
- Collaborated closely with a project mentor to refine goals, validate assumptions, and incorporate **real-world AI considerations** into the solution.

PROJECTS

AniDis - Full-Stack Anime Platform | Next.js, React, TypeScript, FastAPI, PostgreSQL, Redis, Docker, OpenTofu

- Built **AniDis**, an anime discovery and discussion platform.
- Designed FastAPI/PostgreSQL APIs for discovery, profiles, episode progress, follows, threads, and nested comments, plus an auth stack with password, OTP, and OAuth flows.

- Delivered reliable reply notifications via a transactional-outbox publisher, a notification service, and an in-app inbox.
- Migrated the anime catalog to provider-neutral identifiers, eliminating duplicate writes with atomic upserts and cutting query cost via indexed seasonal rows.
- Automated releases with changesets and SHA-tagged images, plus preview environments gated by Cloudflare Access with isolated databases.
- Provisioned DNS and routing in OpenTofu, running Docker Compose stacks behind Traefik with Doppler-synced secrets, OpenTelemetry, and Grafana dashboards.